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Volunteer associations in the Internet age: Ecological approach to understanding collective action

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ABSTRACT
This study draws on theories of collective action and organizational ecology to investigate a particular type of voluntary association: mixed-mode groups. Mixed-mode groups are created and organized online to meet physically in geographically defined ways. An online survey was conducted with 171 randomly sampled groups on Meetup.com—a website that facilitates the creation and coordination of mixed-mode groups. Analysis shows that even when a mixed-mode group implements strategies focused on internal group processes, it benefits from tapping into its external networks to obtain resources for group operation. Also, the differential impacts of internal and external strategies indicate that in the case of environment-prone mixed-mode groups, implementation of internal strategies alone, with or without solicited external networking resources, is helpful for generating positive group outcomes. Together, these results establish that boundary spanning represents not only a type of strategic action to obtain resources and produce positive outcomes but also an inherent and defining mechanism of contemporary voluntary groups for engaging in collective action.

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Boundary spanning; collective action; ecology; Internet; Meetup.com; voluntary associations

As the digital era dawned, many analysts and scholars initially anticipated that new developments in communication, most particularly the Internet, would destroy or severely erode traditional forms of voluntary associations (Nie 2001; Putnam 2000; Shah, Kwak, and Holbert 2001; Turkle 1996). What was underestimated was the possibility of the Internet actually extending the reach of voluntary associations and creating many opportunities for new ones to come into being (Katz and Rice 2002; Wellman et al. 2001). A 2011 Pew Internet & American Life Project report underscores the importance of the Internet in the life of voluntary associations. It reports that 59% of the surveyed Americans think the Internet has played a role in their groups' ability to impact society at large, and 49% believe that their groups can influence local communities because of the Internet (Rainie, Purcell, and Smith 2011).

By engaging in collective action, voluntary associations not only provide benefits to their members but also impact the wider public (Babchuk and Edwards 1965; Hooghe 2003; Putnam 1995; 2000; Stolle 2000; Tocqueville 1968; Wollebæk and Selle 2002). It is therefore important to understand how their sustainability and effectiveness for collective action can be enhanced in the world of the Internet.

Some theorists assert the importance of face-to-face engagement in collective action (Calhoun 1986; Putnam 2000; Sampson 1988; Verba, Schlozman, and Brady 1995). Others contend that people are more empowered to participate in online collective action of relative anonymous interaction and fluid membership (Fulk et al. 1996; Lea and Spears 1991; Postmes, and Brunsting 2002). More recently, new thinking has emerged that instead of holding an either-or (online/offline) lens, we should be concerned with how technology enables varying forms of collective action (Bimber, Flanagin, and Stohl 2012; Flanagin, Stohl, and Bimber 2006).

Although the new conceptualization provides a more nuanced approach for studying fluid forms of contemporary collective action, it also gives rise to the question of where the "collective" lies, that is, who are involved in and affected by the outcomes of collective activity. Organizational ecology theories, on the other hand, characterized by multilevel analysis of the interaction of collective behaviors and environmental influences (Aldrich 1999; Monge, Heise, and Margolin 2008), can provide insights

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into the adaptive process of individuals and the collective and the resulting effect of such adaptive efforts. They can therefore be extended to analyze processes and outcomes of voluntary associations in a more meaningful way.

This study uses a new form of voluntary association, *mixed-mode groups*, to investigate empirically practical and theoretical concerns about contemporary collective action. Simply defined, mixed-mode groups are an emerging type of voluntary association that are created and organized online to interact physically in geographically defined ways (Lai 2014a; 2014b). It looks at groups on Meetup.com—a website that facilitates the creation and coordination of mixed-mode groups.

This article is organized as follows. The next section reviews theories of collective action and ecological approaches and advances hypotheses. Subsequent sections discuss the methodology, results, and conclusions.

Theories of collective action and organizational ecology

General theories of collective action seek to explain the motivation of groups to cooperate and to perform collective action (Olson 1965). In essence, collective action is driven by mutual interests and benefits from coordination action (Marwell and Oliver 1993) and can be seen as a communicative process "insofar as it entails efforts by people to cross boundaries by expressing or acting on an individual (i.e., private) interest in a way that is observable to others (i.e., public)" (Flanagin, Stohl, and Bimber 2006, p. 32). In particular, public goods theory is concerned with the explanation of collective human action to create public goods. Public goods are the outcomes of collective actions taken by two or more people that benefit the general public, some of whom do not contribute to the creation of the public good (Samuelson 1954). Public goods are nonrival—that is, one's use of the public good does not reduce the amount that others can use. Public goods can be tangible (e.g., parks) and intangible (e.g., electronic databases).

Research on collective action generally falls into two categories: generative mechanisms of collective action (mobilization) and the adoption of innovations (Monge and Contractor 2003). The former is centered on the individual and network mechanisms conducive to collective action and the creation of the public good. For example, with individuals and organizations embedded in dense and central networks, public goods are likely to be produced (Laumann, Knoke, and Kim 1985; Monge et al. 1998; Wasko, Teigland, and Faraj 2009). The latter is concerned with the individual variables predicting the adoption and use of innovations within an organizing context. For example, individuals are influenced by their perceptions of gains in deciding their contributions to the information commons such as intranets (Fulk et al. 2004; Wasko and Faraj 2005). They are also influenced by others in terms of adopting a new communication system in organizations (Rice et al. 1990; Yuan et al. 2005). A widely known extension of the public goods theory to technological contexts is Fulk et al.'s (1996) conception of connective and communal goods. Connective goods refer to the functionality of communication and information systems (e.g., intranets and online discussion boards) that allow direct connections between members within the collective, whereas communal goods are the collective information stored and shared among members.

Mixed-mode groups

Mixed-mode groups are created and organized online to interact physically in geographically defined ways (Lai 2014a; 2014b). Mixed-mode groups represent contemporary forms of voluntary associations, but with relatively low thresholds for initiating and participating in associational activities and the flexibility of organizing and interacting in a multimodal way (Bimber, Flanagin, and Stohl 2012). As such, they fulfill ongoing pluralistic needs and interests in human society. Examples of websites that enable mixed-mode groups include Facebook, Craigslist, Meeetup.com, and Twitter. An argument could be made that usage of Facebook and Twitter is diverse compared to Meetup.com, whose primary focus is the creation and organization of groups (based on shared interests and locations) that meet in a physical space. A group created and organized through Meetup.com that engages in hiking activities in the San Francisco area is an example. To more precisely examine how mixed-mode groups traverse online and physical spaces in engaging in collective action, this study focuses on groups organized on Meetup.com.

Following these theories of collective action and public goods, collective action of mixed-mode groups is exhibited through shared information and connections built online (Fulk et al. 1996), as well as through social interactions taking place during face-to-face group meetings (Putnam 2000). Hence, outcomes of this collective action create the public good of a relatively open online space made available to any current and potential members for operation and organizing of the group. What is less known is that outcomes also take the form of impure public goods as the facilitation of personal relationships and shared collaboration is made available only to members who attend the face-to-face meetings. Impure public goods, also called club goods, are similar to public goods because the consumption of club goods is nonrival to the extent that all participating members can have access to the benefits; unlike public goods, club goods have the characteristic of exclusion based on active membership (Sandler and Tschirhart 1980). For example, the activities that a sports team engages in are only available to the attending members; however, within the team, each member is offered equal enjoyment.

Two conceptual challenges stem from this application. First, depending on the space where mixed-mode groups' collective action takes place, the collective goods may take either public or impure public forms. Hence current theorizing about online connective and communal public goods (Fulk et al. 1996) and club goods (Sandler and Tschirhart 1980) appears inadequate or ambivalent in seeking to comprehensively account for the collective action of mixed-mode groups. Second, extant theories and research of collective action are heavily reliant on the production of public goods, which leaves the issues of how the public goods facilitate group outcomes relatively unexplored (Monge et al. 1998). For instance, as part of in-person meetings, mixed-mode groups may visit local restaurants, which helps keep the business open or even helps community overall. It is possible that the public goods of those groups may not only help mixed-mode groups achieve member satisfaction but also spill over to the geographic community. To address these challenges, we argue that ecological theories may serve as a more useful framework for understanding contemporary forms of collective action.

Ecological approaches and collective action

In the organizational context, the ecological perspectives focus on the relationships between organizations and environments and examine the process of how communities and the populations of organizations that constitute them struggle to acquire resources in order to survive, by means of interacting with other members as well as with their environments (Campbell 1965; Hannan and Freeman 1977; McPherson 1983). Central to the survival of organizations are the seemingly mutually exclusive mechanisms: strategic choice and environmental forces. Simply defined, voluntarism (choice) and determinism are the two categories that describe the preponderance of the organization and the environment in deciding organization decision making, change, and adaptation (Hannan and Freeman 1977). A broadly accepted view is that these two categories should be seen as interacting with each other, which in turn results in organizational behaviors (Baum and Shipilov 2006).

The advantage of the ecological perspectives lies in their ability to explain phenomena using the same theoretical process at different levels (Monge, Heise, and Margolin 2008). In contrast with the research on collective action and public goods that treats organizations as relatively closed systems, ecological theories see individuals and organizations subject to influences from the environment (McPherson 1983). In the collective action realm, ecological theories can thus be used to understand how individuals within organizations, as well as organizations, adapt themselves by coping with internal and external dynamics in order to accomplish collective action. These adaptations can take the form of strategic actions such as involving external social actors to secure necessary resources for group operation and task completion (Ancona 1990; Ancona and Caldwell 1988; 1992). Moreover, examining outcomes of voluntary associations reflects the ecological notion of *co-evolution*, which concerns the interactions between the organization and its environment, and the subsequent consequences of these interactions for the environment (Baum and Singh 1994).

Instead of isolating online or offline environments, an ecological view can provide a systematic analysis of how mixed-mode groups incorporate both environments in producing outcomes of collective action, and how these interactions with the environments in turn influence the groups as well as the environments. To answer these inquiries, and to inform the development of hypotheses (and in light of the lack of prior research linking strategic action and environmental influences in the context of technology-enabled collective action), we draw on existing research that addresses group boundary spanning. An emphasis on group boundary spanning provides a conceptual approach that fits the context of this study and constitutes an ecological lens for examining groups' interaction with the environment and the subsequent outcomes of such interaction.

Enactment of strategies within and across boundaries

Research on group boundary spanning (Ancona 1990; Ancona and Caldwell 1988) has great bearing on the ecological perspective because it examines how groups engage in communication activity involving external social actors in order to secure necessary resources for group operation and task completion. From the group boundary-spanning viewpoint, individual mixed-mode groups cannot avoid interacting with their environment, which consists of the physical community (e.g., local businesses and organizations) and other similar and dissimilar mixed-mode groups. Rooted in resource dependence theory (Pfeffer and Salancik 1978), boundary-spanning research suggests that groups that can better manage their degree of dependence on their environments and related boundary-spanning activities can perform better than those that focus only on internal dynamics (Ancona and Caldwell 1988).

A set of strategies has been identified as representing the patterns of external activities groups perform, including ambassadorial efforts (persuading others to support groups and lobbying for resources), task coordination (coordinating and negotiating with outsiders for technical or design issues and obtaining feedback), scouting missions (general information gathering and mapping and scanning the environment), and guard responsibilities (avoiding releasing information) (Ancona and Caldwell 1992).

Nonetheless, the aforementioned benefits of boundary spanning activity do not minimize the equally important role of internal activity in influencing group outcomes. Despite contingencies, such as temporal development of a group or environmental influences, maintaining a balance between internal and external activity is critical to group outcomes (Ancona and Caldwell 1988; Choi 2002). Studies have identified the positive influence of effective internal and external strategies on group outcomes (Ancona and Caldwell 1988; 1992; Druskat and Wheeler 2003; Faraj and Yan 2009; Guinan, Cooprider, and Faraj 1998). But it must be noted that boundaries of voluntary associations are often blurred and porous (Aldrich 1999) and associations typically lack sufficient resource bases (e.g., personnel, leadership) and structural complexity to engage in externally reciprocated exchange relations and to obtain external information and support, which in turn can influence overall effectiveness (Knoke and Prensky 1984). Therefore, it is both a theoretical and practical test to apply the existing research on group boundary spanning to examine the accomplishment of collective action by mixed-mode groups, a type of voluntary association. In light of the preceding discussion, the following set of hypotheses examines the relationship between the strategies that mixed-mode groups use and their resulting outcomes:

H1a: The more strategies a mixed-mode group uses involving internal group processes, the more likely it is that a group will perceive the accomplishment of positive group outcomes.

H1b: The more strategies a mixed-mode group uses involving external actors, the more likely it is that a group will perceive the accomplishment of positive group outcomes.

Networks and resources

According to the perspective of group boundary spanning, resource acquisition is the central mechanism that drives boundary spanning activity and external networking. In conceptualizing external group interactions, two approaches have generally been adopted: a qualitative or descriptive approach and a network analytical approach (Joshi, Pandey, and Han 2009). The former provides insights into the processes of how groups initiate strategies to enhance external visibility, secure information, and coordinate with others outside the group (e.g., Ancona 1990), while the latter largely draws on social capital theory to investigate how groups invest in external relationships that create values and result in resources (e.g., Oh, Chung, and Labianca 2004). Social capital refers to the resources embedded within and available through social relations that are accessed and/or mobilized by actors in purposive actions (Coleman 1988; Lin 1999; Nahapiet and Ghoshal 1998). Another similar concept, which is often examined in the interorganizational context, is social embeddedness. Research on social embeddedness is primarily concerned with how microsocial factors, such as an employee's social networks, influence the performance of a firm (Granovetter 1985).

Incorporating the notions of social capital and social embeddedness in the ecological framework can help shed light on how organizations and groups secure resources and maintain existence by tapping into their social networks. For example, Westphal, Boivie, and Chng's (2006) study exemplifies how social embeddedness, in the form of friendship ties maintained among corporate leaders, provides advantages for organizations to manage resource dependence. Comparably, empirical studies on work groups show that groups that more frequently secure resources with different people outside those groups tend to attain better outcomes, such as getting higher performance ratings, producing more innovations, and having more efficient project completion (e.g., Hansen 1999; Tsai 2001). Oh, Chung, and Labianca's (2004) findings indicate that groups can attain maximum effectiveness by investing in intragroup closure relationships and expansive intergroup relational structures, through which resources can be accessed. Nonetheless, extant research is mostly developed and tested on work groups; less is known about whether these arguments of network resources can apply to voluntary associations and mixed-mode groups. Hence, a second set of hypotheses is proposed to investigate the possibility of engaging in external communication to obtain resources for mixed-mode groups:

H2a: The more a mixed-mode group is embedded in its external network contacts, the more likely it is that a group will receive resources necessary for group operation.

H2b: The more a mixed-mode group receives resources necessary for group operation, the more likely it is that a group will perceive the accomplishment of positive group outcomes.

While there has been extensive discussion on the advantages of resource networks in facilitating positive organizational outcomes, the ecological perspective emphasizes the strategic choices enacted on by organizations as they explore the networking environment and decide whom to connect with (Aldrich and Pfeffer 1976; Monge, Heise, and Margolin 2008). At first glance, it might seem reasonable that only when enacting external strategies will a mixed-mode group feel the need to engage in external communication. Research has shown that organizations achieving high performance often have a wide range of external strategies on hand, for example, initiating multiple, simultaneous ties with potential outside partners (Ozcan and Eisenhardt 2009). Yet it is also plausible that even if a group focuses on internal strategies, it may acquire resources through external communication. For instance, a group might evaluate other groups' situations when implementing a group policy or deciding upon an appropriate activity format. Hence, another two hypotheses are proposed to examine the relationships among scopes of strategies, network embeddedness, and the resources received:

H3a: Implementing more internal and external strategies helps a mixed-mode group to become embedded in its external network contacts.

H3b: Implementing more internal and external strategies helps a mixed-mode group to receive resources necessary for group operation.

The review thus far has revealed a common observation that a group is capable of enacting external and internal strategies, building on the networking benefits it has accumulated, which in turn helps achieve group outcomes. A well-researched line of work has identified the advantages of both internal and external network links in facilitating group outcomes (e.g., Oh, Chung, and Labianca 2004; Reagans, Zuckerman, and McEvily 2004). Yet there is a limited understanding of whether, and how, network communication and resource acquisition might enhance the effects of internal and external strategies on group outcomes. Ancona's (1990) study indeed points out the accentuated effects of boundary spanning on group performance occurring in groups that need outside resources, support, or information. Compared to purely online groups, the sustainability of mixed-mode groups is closely linked to the factors outside the group, such as the existence of other similar groups or availability of partnering organizations in the local area (Lai 2014a). It is thus a possible conjecture that mixed-mode groups' receipt of resources from external contacts enhances group efforts invested in external strategies, more than in internal strategies, when it comes to producing outcomes. It is therefore hypothesized that mixed-mode groups' use of certain types of strategies will exhibit different effects on group outcomes through network communication and resource acquisition:

H4: Compared with internal strategies, implementing external strategies is more likely to facilitate a mixed-mode group's accomplishment of positive group outcomes when a group is embedded in its external network contacts and acquires resources.

Methods

The data were collected from Meetup.com, a website specifically designed to facilitate the creation of online groups based on shared interests and physical locations and to coordinate offline group meetings. Moreover, Meetup.com fits the needs of this study because of its large user base (over 238,000 local groups, 25.5 million users). A stratified random sampling strategy was used to generate 2,000 Meetup groups. Online survey respondents were recruited from this sample. Considering the importance of age in group survival, as observed in a previous analysis by the author (Lai 2014a),[1] the entire population of Meetup groups (mainly in North America) was sorted into three subpopulations based on group age (less than 1 year, between 1 and 2 years, and more than 2 years). The 2,000 groups were then randomly pulled from these subpopulations in proportion to the age distribution: 48.4% (968 groups) aged less than 1 year, 24.73% (495 groups) aged between 1 and 2 years, and 26.87% (537 groups) aged more than 2 years.[2] In situations where the same organizers managed multiple groups, only one group was randomly selected for invitation. Moreover, some groups listed in the sampling frame disbanded by the time the invitation was extended. Consequently, 1,237 of the 2,000 sampled groups were considered as valid subjects for invitation.

Those 1,237 selected organizers were asked individually through the contact function on Meetup.com to participate in the online survey, and 171 responded (response rate = 13.82%).[3] Although this response rate was disappointing, it was not uncommon to experience comparable low rates in online surveys (Dillman 2000). Data collection lasted approximately 2 months, from December 4, 2010, to February 1, 2011. The age distribution of the participating groups was slightly different from the Meetup population. The groups aged less than 1 year were 29.76% of the sample (underrepresented), those aged between 1 and 2 years were 30.36% of the sample, and those over 2 years were 39.88% of the sample (overrepresented). Moreover, among the 171 groups, about one-third (31.8%, $n = 54$) of the groups experienced a leadership change as the originator of the group yielded the running of the group to someone else.

Instruments and measures

To test the hypotheses, this study evaluates different sets of variables, including group strategies, group impacts, network embeddedness, and resource acquisition (see Figure 1). Because of the study's exploratory nature, these variables were largely informed by the literature of group boundary spanning and network theory (e.g., Ancona and Caldwell 1992; Oh, Chung, and Labianca 2004). Yet to increase the relevance of the measures, the previously conducted interviews with Meetup group organizers were used to modify the scales to make them relevant and feasible for the Meetup groups under study (Lai 2014a; 2014b). To ensure content validity, the survey was also pilot tested with five Meetup organizers to validate original scales and clarify question wording.

Types and scope of strategies used

To examine the patterns of internal and external activities that a group engages in, we looked at the scope of strategies they used. They were categorized as internal and external strategies. Each respondent was asked to indicate whether and how often he or she used a given strategy to run the group, using a 5-point scale (1 = *never*, 2 = *once*, 3 = *a few times*, 4 = *many times*, and 5 = *regularly*). Internal strategies were measured using 10 items and external strategies using eight items based on the examples provided by the organizers during the previously conducted interviews. These 10 internal strategies were as follows: group policy, requirement of member dues, member involvement, diversity of activities, creation of subgroups, private events, focused topics, use of technology for communication, diversity of locations, and regular events. These eight external strategies were as follows: copy other Meetup groups, copy other non-Meetup groups, cross-post events by other Meetup groups, cross-post events by other non-Meetup groups, joint events with other Meetup groups, joint events with other non-Meetup groups, activity as part of local events, and interaction with local venues. Each of the 18 items was further dichotomized into "0," where the given strategy is never used, and "1," where the strategy is used at least once. Index values ranged from 0 to 10 for internal strategies and from 0 to 8 for external strategies. Two separate indexes were then created by aggregating the values of the 10 items for internal strategies ($M = 5.967$, $SD = 1.986$) and eight items for external strategies ($M = 3.831$, $SD = 2.365$).

Network embeddedness

Directionality was considered when measuring a group's embeddedness with its external network contacts through communication. To measure incoming flow of networking, respondents were asked to indicate how often other Meetup groups and other non-Meetup groups or organizations have contacted them over the

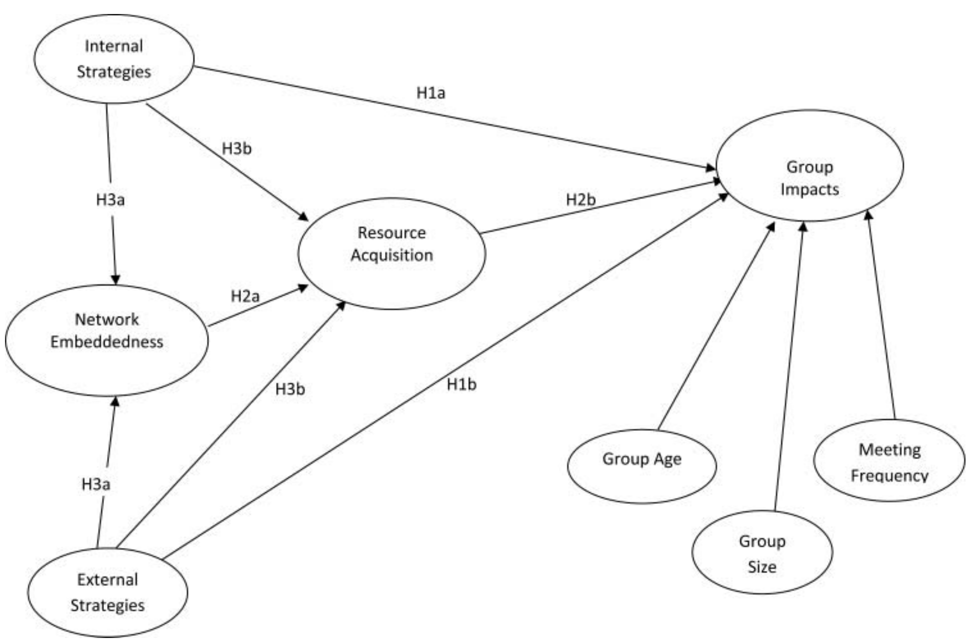

Figure 1. The conceptual model.

course of their groups' development, using a 6-point scale (1 = *never*, 2 = *once or twice a year*, 3 = *once or twice a month*, 4 = *once or twice a week*, 5 = *several times a week*, 6 = *nearly every day*). Another set of items examined a group's outgoing networking with five different contacts, including other Meetup groups, local venues to hold group events, other organizations, personal contacts, and members of other Meetup groups. These items were also assessed on a 6-point scale with response options ranging from 1 = *never* to 6 = *nearly every day*. The aggregated latent variable of network embeddedness ($\alpha = .806$, $M = 1.907$, $SD = 0.653$) consisted of these six items that measured the frequency of a group's overall communication with its external network contacts.[4]

Resource acquisition

Inspired by boundary-spanning literature that examines a network's internal density, external range, and impacts on group performance (e.g., Reagans, Zuckerman, and McEvily 2004), resource acquisition was investigated in two dimensions: density and diversity of resources received. Respondents were asked to indicate how often they had received help of any sort for running the group from six types of network contacts using a 6-point scale (1 = *never*, 2 = *once or twice a year*, 3 = *once or twice a month*, 4 = *once or twice a week*, 5 = *several times a week*, 6 = *nearly every day*). The aggregated latent variable of density of resources received ($\alpha = .796$, $M = 1.722$, $SD = 0.624$) consisted of six items that measured the frequency of a group's receipt of resources from its network contacts.[5] Each of the six items was further dichotomized into "0," where the given contact was never sought after for support, and "1," where the contact provided support at least once.[6] The index of diversity of resource received was then created by aggregating the values of the six items and the index value ranges from 0 to 6 ($M = 2.919$, $SD = 1.857$). To avoid the potential problem of multicollinearity, a two-stage approach was performed to create a second-order construct called *resource acquisition* (Agarwal and Karahanna 2000; Chin and Gopal 1995; Osei-Bryson, Dong, and Ngwenyama 2008). The latent variable scores of density of resources received were initially estimated, and these scores, along with diversity of resources, were then entered as indicators for the second-order construct of resource acquisition ($M = 2.021$, $SD = .899$). Resource acquisition was the variable used in the assessment of the complete model.

Group impacts

Accomplishment of positive group outcomes was measured through the concept of group impacts, which captures internal and external outcomes (Smith 2000) and thus reflects the co-evolution between groups and the environment (Baum and Singh 1994). Informed by Meetup organizers' responses during the previously conducted interviews (Lai 2014a 2014b), a group's impact was measured using six items that assessed aspects of interpersonal relationships, community development, and local interaction. These dimensions coincided with the internal and external impacts of voluntary associations as suggested by Smith (2000). Six items were assessed on a 5-point scale, with response options ranging from 1 = *strongly disagree* to 5 = *strongly agree*. Some examples of the possible responses include "my group has contributed to building and/or maintaining relationships among members" and "my group has maintained local area interest in the group topic" ($\alpha = .874$, $M = 4.132$, $SD = 0.756$).

Control variables

Three additional variables were included as controls: *group age*, *group size*, and *meeting frequency*. It has been argued that a voluntary association's impacts are usually observable after 1 or 2 years of its establishment (Smith 2000). In the boundary-spanning literature, temporal development is suggested to relate to boundary-spanning activity, which in turn affects group outcomes (e.g., Sawyer, Guinan, and Cooprider 2010). Hence, group age was included as a control variable in the model. Note that at the sampling stage, group age was considered a salient factor pertaining to group outcomes, and thus stratified sampling was implemented to ensure representativeness of groups of different ages. It is logical that group size and frequency of face-to-face interaction are also key factors to generate impacts of voluntary associations: A group with enough members who frequently show up for activities would be likely to see significant impacts (Smith 1999). Interestingly, research thus far has found mixed yet significant effects of face-to-face meetings on the continued collective action of online groups (e.g., Sessions 2010; Shen and Cage 2015). Respondents were asked to provide their groups' month and year of establishment on Meetup.com. Based on this information, the group ages for all 171 groups were calculated to the cut-off date of February 1, 2011. Group size was based on the respondents' self-reported numbers of members at the time of the survey. Respondents were also asked to identify how often (number of times) their groups met in the last month, to assess a group's meeting frequency.

Analysis procedures

Given the exploratory nature of this study, partial least squares (PLS) path modeling was used (see reviews of PLS modeling by Henseler, Ringle, and Sinkovics [2009] and Sosik, Kahai, and Piovoso [2009]). Data were analyzed using Smart PLS (Ringle, Wende, and Will 2005), an easy-to-use PLS path modeling software (Temme, Kreis, and

Hildebrandt 2010). According to Chin (1998), PLS models can be evaluated using a two-step process: (1) the assessment of the outer/measurement model (i.e., reliability and validity of reflective constructs) and (2) the assessment of the inner/structural model (i.e., variance explanation of dependent variables, effect sizes, and significance of the path coefficients). In the following section, the measurement model is assessed using these criteria, after which the results of assessing the structural model and the significant paths are reported.

Results

A measurement model is typically assessed based on its reliability and validity (Henseler, Ringle, and Sinkovics 2009). Except for network embeddedness and group impacts, the other six variables were measured with a single indicator or an aggregated index and thus were not included in this part of the assessment. For the assessment of reliability, outer loadings of all indicators for network embeddedness and group impacts were greater than .60, and the composite reliability and Cronbach's alpha for these two latent variables were well above .70, indicating the measurements were reliable (Bagozzi and Youjae 1988; Fornell and Larcker 1981; Gotz, Liehr-Gobbers, and Krafft 2009; Sosik, Kahai, and Piovoso 2009) (see Table 2, shown later). There was also evidence of sufficient convergent and discriminant validity.

First, the average variance extracted (AVE) for network embeddedness and group impacts exceeded the recommended criterion of .50, suggesting sufficient convergent validity because these two latent constructs can explain at least 50% of their indicators' variances on average (Chin 1998; Fornell and Larcker 1981; Gotz, Liehr-Gobbers, and Krafft 2009). Second, the AVE values of network embeddedness and group impacts were greater than their squared correlations with any other constructs, meaning each latent variable shared more variance with its own assigned indicators than with another latent variable representing a different block of indicators (Chin 1998; Fornell and Larcker 1981; Gefen, Straub, and Boudreau 2000; Gotz, Liehr-Gobbers, and Krafft 2009) (see Tables 1 and 2). Another criterion of discriminant validity is to check whether the loading of each indicator is greater than all of its cross-loadings (Chin 1998; Gefen, Straub, and Boudreau 2000). A cross-loadings table (see Table 3) revealed that each item loading was higher on its assigned construct than on the other constructs, supporting adequate discriminant validity. All of the t values of outer loadings were greater than 2.58 ($p < .01$).

In terms of the overall model fit of the structural component, three criteria were used (Chin 1998; Henseler, Ringle, and Sinkovics 2009). First, Smart PLS generated the results of R-squared for the three endogenous variables: group impacts ($R^2 = .234$), resource acquisition ($R^2 = .470$), and network embeddedness ($R^2 = .428$). In other words, 23% of the variance of group impacts, 47% of the variance of resource acquisition, and 43% of the variance of network embeddedness, respectively, were explained by the model. These values approximate or exceed the criterion of $R^2 = .26$ (Cohen 1988) for large effect sizes, which supports the argument that the conceptual model offers an adequate explanation of the analytical results.

Second, an F test was conducted to assess whether the model is a significant overall fit to the data. The results showed that the four predictors together (internal strategies, external strategies, network embeddedness, and resource acquisition) had a substantive effect on the endogenous

Table 1. Summary of intercorrelations among the study variables.

Variables	1	2	3	4	5	6	7	8	9	10
1. Group age	—									
2. Group size	.548**	—								
3. Meeting frequency	.035	.043	—							
4. Internal strategies	.177*	.308**	.271**	—						
5. External strategies	.183*	.344**	.161*	.457**	—					
6. Network embeddedness	.144†	.400**	.185*	.442**	.630**	—				
7. Density of resources	.076	.365**	.153*	.342**	.529**	.687**	—			
8. Diversity of resources	.097	.252**	.108	.364**	.489**	.595**	.861**	—		
9. Resources (second-order)	.090	.320**	.136†	.366**	.528**	.665**	.965***	.964***	—	
10. Group impacts	.189**	.178*	.229**	.396**	.318**	.302**	.264**	.266**	.275**	—
Number of items	1[a]	1[a]	1[a]	1[b]	1[b]	6	6	1[c]	2[d]	6
Mean	25.214	219.036	3.275	5.967	3.831	1.907	1.722	2.919	2.021	4.132
SD	20.532	289.67	4.138	1.986	2.365	.653	.624	1.857	.899	.756

Note. Total number of participants studied, $N = 171$.
[a]Group age = one item measuring observed group age, group size = one item measuring observed membership size, meeting frequency = one item measuring observed meeting frequency.
[b]Internal strategies = the sum of 10 items; external strategies = the sum of eight items.
[c]Diversity of resources = the sum of six items measuring occurrence of receiving resources from contacts.
[d]Resource acquisition is a second-order construct consisting of density and diversity of resources received.
†$p < .10$; *$p < .05$; **$p < .01$; ***$p < .001$.

Table 2. Reliability and validity of multi-indicator latent variables.

Construct	AVE	Composite reliability	Cronbach's alpha	Indicator	Mean	*SD*	Loading	*T* Value
Network embeddedness	.509	.861	.806	NE1	1.88	.918	0.663	11.230
				NE2	2.12	.883	0.623	11.473
				NE3	1.70	.877	0.802	23.732
				NE4	2.09	.973	0.681	10.355
				NE5	1.82	.924	0.796	23.096
				NE6	1.88	1.115	0.700	14.269
Density of resources[a]	.500	.854	.796					
				DR1	2.48	1.035	0.502	5.673
				DR2	1.44	.714	0.797	17.161
				DR3	1.76	.996	0.605	6.415
				DR4	1.55	.834	0.796	17.101
				DR5	1.96	1.069	0.717	10.690
				DR6	1.63	.914	0.776	14.317
Resources (second-order)	.930	.964	.925					
				DeR[b]	1.722	.624	0.968	162.951
				DiR	2.919	1.857	0.961	102.807
Group impacts	.609	.903	.874					
				GI1	4.15	1.013	0.768	15.845
				GI2	4.55	.790	0.781	14.663
				GI3	4.44	.919	0.763	14.986
				GI4	4.06	1.059	0.792	14.300
				GI5	3.92	1.031	0.826	22.064
				GI6	3.39	1.262	0.752	18.378

[a]Factor loadings of the indicators of density of resources were calculated separately in the initial model when the second-order construct of resources was not present.
[b]DeR = density of resources, DiR = diversity of resources.

variable of group impacts ($F[4,163] = 6.756, p < .001$). Last, a global criterion of goodness-of-fit for PLS path modeling, the GoF index, was used. The GoF index is the geometric mean of the average communality index (outer measurement model) and the average R^2 value of the endogenous latent variables (Tenenhaus et al. 2005). It ranges from 0 to 1, where higher values represent better path-model estimations (Henseler, Ringle, and Sinkovics 2009). The GoF value of the model was 0.614, which exceeds the cutoff value of 0.36 for large effect sizes of R^2 (Wetzels, Odekerken-Schröder, and van Oppen 2009).[7] In sum, the results indicate a good prediction performance of the model overall.[8]

Hypothesis testing

Hypotheses were tested by examining the significance of the path coefficients through asymptotic t statistics, which were obtained by bootstrapping resampling (500 samples) (Chin 1998).[9] Figure 2 presents the estimates obtained from the PLS analysis. Results showed that the paths from the two control variables (group age and meeting frequency) significantly predicted group impacts. Both internal and external strategies significantly predicted network communication, which in turn affected resource acquisition. As such, H2a and H3a

Table 3. Factor loadings and cross loadings of measures.

	Network embeddedness	Group impacts	Resources	Group age	Group size	Meeting frequency	Internal strategies	External strategies
NE1	*0.663*	0.259	0.405	0.165	0.269	−0.020	0.290	0.406
NE2	*0.623*	0.301	0.351	0.113	0.263	0.171	0.395	0.336
NE3	*0.802*	0.145	0.532	0.120	0.303	0.157	0.252	0.546
NE4	*0.681*	0.380	0.436	0.105	0.300	0.179	0.404	0.388
NE5	*0.796*	0.263	0.542	0.088	0.292	0.183	0.340	0.572
NE6	*0.700*	0.041	0.567	0.027	0.290	0.113	0.238	0.413
GI1	0.157	*0.768*	0.248	0.183	0.181	0.152	0.317	0.195
GI2	0.156	*0.781*	0.242	0.061	0.139	0.161	0.250	0.215
GI3	0.208	*0.763*	0.221	0.000	0.045	0.176	0.260	0.213
GI4	0.266	*0.792*	0.232	0.204	0.120	0.199	0.345	0.220
GI5	0.274	*0.826*	0.138	0.151	0.165	0.185	0.301	0.282
GI6	0.337	*0.752*	0.215	0.293	0.185	0.205	0.394	0.372
DeR	0.691	0.265	*0.968*	0.076	0.365	0.153	0.342	0.529
DiR	0.598	0.270	*0.961*	0.097	0.252	0.108	0.364	0.489

Note. Italicized loadings explain that items (NE1-NE6,GI1-GI6, & DeR, DiR) have higher loadings with their respective constructs (network embeddedness, group impacts, resources) than with other constructs.

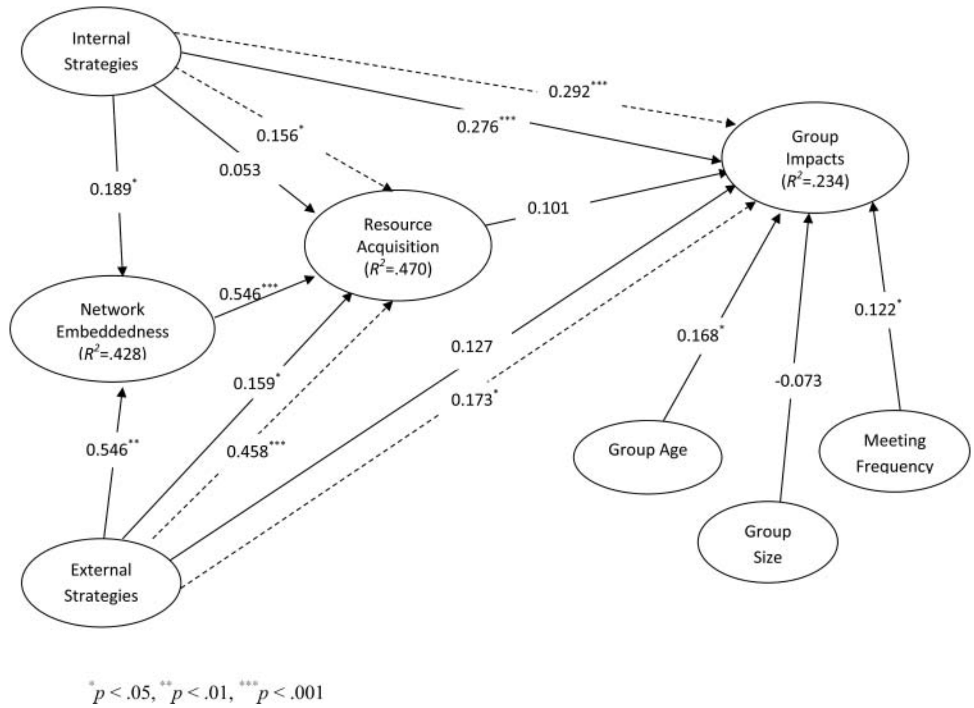

Figure 2. The resulting model via PLS analysis. The path coefficients displayed are standardized. Dashed lines refer to the total effects. Significance: $^{*}p < .05$, $^{**}p < .01$, $^{***}p < .001$.

were supported. Network embeddedness was useful for groups in acquiring resources when implementing either internal or external strategies. Yet the external and internal strategies differed, as the former had both significant direct and total effects on resource acquisition, while the latter only had significant effects on resource acquisition after including network embeddedness ($\beta = .156$, $t = 2.257$, $p < .05$). Thus, H3b was partially supported. Both internal and external strategies had significant total effects on group impacts, but only internal strategies had direct effects on group impacts. External strategies, on the other hand, had effects on group impacts after including network communication and resource acquisition ($\beta = .173$, $t = 2.292$, $p < .05$). Based on these results, it appears that H1a and H4 were supported while H1b was partially supported.[10] Nonetheless, the analysis did not show a positive effect of resource acquisition on group impacts, and thus H2b was not supported.

Discussion

In terms of accounting for multimodal collective goods and inevitable group interactions with multimodal environments, existing theories of collective action and public goods have limited applicability when it comes to mixed-mode groups. To address this limitation, this study employs ecological theories to investigate the process and outcomes of contemporary collective action. Groups from the website Meetup.com were studied as an example of mixed-mode groups. While being embedded in the environment consisting of individuals and groups within and outside Meetup.com, online and offline, these Meetup groups enacted strategies coping with internal and external dynamics. Especially, interacting with the environment in the form of external networking helped these groups acquire resources necessary for group operation and generate impacts that influenced members and the local community.

Ecological perspectives provide insights into the importance and usefulness of Internet use in facilitating group outcomes. We see that the Internet, along with face-to-face communication, provides a means for mixed-mode groups to initiate and maintain communication with external social actors as part of the group strategy for securing resources. It thus makes theoretical sense to conclude that the Internet has become incorporated in the adapted organizational practices of mixed-mode groups and that group impacts are likely accomplished through these efforts.

Relationships between mixed-mode groups and the environment

This study extends the group boundary spanning literature to voluntary associations, in particular to the mixed-mode group context. Meetup groups, being embedded in their hosting environment of Meetup.com as well as other larger resource environments, tend to have amorphous and porous boundaries that facilitate interaction within and across groups. Groups were observed to engage in external communication with individuals and organizations outside Meetup.com, which helped them to acquire resources necessary for group operation. Unlike task groups, work organizations, or formal nonprofit organizations, which are the dominant targets of study under boundary-spanning and organizational ecological research, mixed-mode groups resemble a type of association that may not be easily defined as small group or organization. In response to Knoke and Prensky's (1984) questioning of the applicability of contemporary organization theories to voluntary associations, this study provides empirical evidence for the usefulness of the ecological lens in explaining and understanding voluntary associations in general and mixed-mode groups in particular.

By considering the broader ecology of voluntary groups in general, and in particular mixed-mode groups, the multiple ways of accomplishing contemporary collective action and interaction with environments can be more clearly understood. Ecological dimensions, though taking different forms, are found to be integral to this process. These are evidenced by the significant direct effect on resource acquisition from external strategies and the significant total effect on resource acquisition from internal strategies executed through network embeddedness. Even when a mixed-mode group implements strategies related to internal group processes, it benefits from tapping into its external networks to obtain resources for group operation. Also, the differential impacts of internal and external strategies indicate that in the case of environment-prone mixed-mode groups, implementation of internal strategies alone, with or without external networking resources solicited, is helpful for generating positive group outcomes. This finding is consistent with research on traditional work groups (e.g., Guinan, Cooprider, and Faraj 1998; Mathieu et al. 2008). Yet these networking resources will be particularly useful for mixed-mode groups, whose environments extend to both online and offline domains, which enable them to accomplish similar outcomes with an adequate set of external strategies configured. Together, these results establish that boundary spanning represents not only a type of strategic action to obtain resources and produce positive outcomes but also an inherent and defining mechanism of contemporary voluntary groups for engaging in collective action.

The dynamics of how mixed-mode groups practice boundary spanning and interact with the multimodal environments merits further research as it bears on questions of environmental opportunities and constraints in human collective behaviors (Baum and Shipilov 2006). Additional research could enrich the utility of ecological theories for understanding contemporary collective action. Environments may merely represent untapped opportunities or even constraints for mixed-mode groups, if no adequate strategic actions are configured to take advantage of them. Within the population of Meetup.com—consisting of tens of thousands of groups—members are likely to belong to multiple groups. Echoing previous research (Ancona and Caldwell 1988), the condition of multiple affiliations is conducive to boundary spanning, which may also explain its beneficial effects on the Meetup groups under study. A few words are in order concerning the absence of a relationship between resource acquisition and group impacts as found in this study since this outcome is relevant to the trajectory of prospective investigations. Due to technical difficulties, survey respondents were only asked about the general resources they received from contacts, without further details collected concerning the specific content and type of resources involved. Moreover, the measurements of network embeddedness and resource acquisition did not differentiate among face-to-face and electronic means. These measurement shortcomings may help explain the nonsignificant relationship between resource acquisition and group impacts. Another possibility is that instead of an aggregate set of resources, certain resources weigh more heavily than others in leading to group impacts.

Future research should include granular inspections of network communication and resource acquisition to better understand how mixed-mode groups interact with their environments. Network links provisioned with different resources have been found to influence each other and together determine the evolution of organizational communities (Lee and Monge 2011). Specifically, studies can pay more attention to the degree of multimodal communication and the resulting group impacts. It is possible that the importance of resource acquisition may vary depending on the level of multimodality. Through more diverse modes of organizing (e.g., Meetup, Facebook, offline), a group is more likely to tap into different generators of resources from its environment, and achieve desired outcomes accordingly.

Incorporating both internal and external dimensions of strategic variations and group outcomes, this study contributes to the existing research on intraorganizational and organizational ecology and evolution. Applying the concept of internal evolution (Miner 1994) and co-evolution (Baum and Singh 1994), organizers can be said to play an important role in deciding which strategies to act on, choose, and retain as part of a group's routine practices. Building on these strategic choices, organizers further adjust groups' relationships with other

Meetup and non-Meetup groups and organizations, which in turn results in the co-evolution of groups and the external environments by producing impacts. Yet the fluid and blurred group boundaries may also render the leadership function less rigid (Flanagin, Stohl, and Bimber 2006). Future research should examine the conceptualization of leadership in the collective action of mixed-mode groups. It is hypothesized that leadership may become more emergent and shared as members can take on the leadership role whenever they show the ability to facilitate the group's online internal coordination as well as interaction with other online groups. Organizers were asked about the strategies they used, but without differentiating online and offline domains, a regrettable limitation of the data-acquisition methodology. For future research, a worthwhile topic of inquiry would be the degree to which groups' strategic actions taking place online and offline are similar or different and how that variation affects the emergence of leadership and group impacts.

Limitations and conclusion

There are several limitations of this study. First, the small-to-moderate sample size and uncertainties over true respondent randomness led to the selection of non-parametric PLS for statistical analysis and recognition that the findings may be interpreted with caution. Second, the measurements of the variables used were drawn mostly from the existing literature and previously conducted interview data. Though this argues for their construct validity, it was not possible to gain independent validation of them in this context. Future efforts should focus on applying the scales to a different population of mixed-mode groups and integrating established scales. Third, due to resource and time constraints, this study could only collect data at the group level, so group organizers were the ones providing responses to the survey questions. Yet in defense of this approach, it was observed that, on behalf of the group, the group organizers were capable of answering the questions related to group development and the strategies used, and they also had the best knowledge of group interaction with external actors. Fourth, a cross-sectional survey limits causality claims, as is the case with all research derived with this method.

Despite these limitations, this study makes theoretical contributions by comparing collective action and ecological theories to answer two important questions in the contemporary society: How is collective action performed by new forms of human associations, and with what effects? This study suggests in the context of a digitally networked society that there are advantages to using ecological theories relative to collective action theories to understand the emerging technologically enabled behaviors at the interstice of organizational processes and human communication. A reason why ecological theories should be applied is because theories of collective action mostly focus on the generation of public goods, especially in either online or face-to-face forms, but not on how public goods facilitate group outcomes.

Given the importance of voluntary associations in society on various levels, these theories cannot explain how public goods actually "do good" in society. In ecological theories, a common understanding is that both choice and determinism are important in influencing organizational behaviors. The findings reported here seem to confirm the utility of the ecological perspectives by showing the interaction of strategic choices and environmental constraints involved in the collective action of Meetup groups, which in turn influenced the group as well as the local community. Hence, in this increasingly important context, theories of organizational ecology are a better framework for understanding collective action because they can account for behaviors at both individual and organizational levels, and for mutual influence between the group and the environment.

Notes

1. More details about the interviews and the findings from the interviews were reported in Lai (2014b).
2. Meetup.com provided the information about the age distribution in Meetup groups and helped generate the sample.
3. Nonresponse bias analysis was conducted by comparing the earliest and latest third of respondents for each of the variables. No significant differences were found between these two groups of respondents on any of the variables.
4. In the network embeddedness, six items (NE1–6) were included. The item of communication with personal contacts was not included because of its low factor loading (.569). Including this item also reduced the AVE of network embeddedness below the threshold of .50.
5. Six network contacts were included: group members, other Meetup groups, local venues, other groups/organizations, personal contacts, and members of other Meetup groups. One of the six items (DR1) had a factor loading below .60 (.502), but it was still included considering the construct validity of this measurement. Moreover, the AVE was still above the threshold of .50 when including this item.
6. A descriptive analysis was conducted. The results showed that groups answering "1 = never" and those answering "2 = once or twice a year" were distinct because the former accounted for a certain percentage of the sampled groups (ranging from 20% to 60%). It means that groups that did not engage in any resource acquisition effort were different than those groups that did so at least once or twice a year. Accordingly, this enhances the validity of our

decision to transform the items into dichotomized ones based on the original options of 1 (never) and 2–6 (once or twice a year to nearly every day).

7. $F = [(R_2^2 - R_1^2)/(k_2 - k_1)]/[(1 - R_2^2)/(N - k_2 - 1)] = [(0.234 - 0.107)/(7 - 3)]/[(1 - 0.234]/171 - 7 - 1] = 6.756$, with [(7−3), (171−7−1)] degrees of freedom. R_2^2 is for the superset model that includes the set of main predictors, R_1^2 is the baseline model, k_2 is the number of predictors for the superset model, k_1 is the number of predictors for the baseline model, and N is the sample size. See Chin (1998). GoF = sqrt(average(AVE) × average(R^2)) = sqrt (0.613 × 0.377) = 0.614 (Tenenhaus et al. 2005).
8. Despite the limitations, the proposed model was run with AMOS to ensure its validity. The results were comparable to those from the analysis with PLS (Smart PLS).
9. Following Chin's (1998) recommendation, we performed bootstrapping using 500 samples. But to verify the results and ensure the stability of the magnitude and significance of the estimated path coefficients, the model was replicated with bootstrap samples of 2,000 and 5,000. The results were consistent across these samples.
10. To further determine the significance of the indirect effects, bootstrapping simple mediation was used with 5,000 bootstrap resamples (Hayes 2009; Preacher and Hayes 2008). Results showed that the indirect effects of both types of strategies on resource acquisition through network embeddedness were significant. Thus, the total effects of external strategies on resource acquisition were ascribed to both direct and indirect effects; in contrast, the total effects of internal strategies on resources were mainly attributed to the indirect effects. Nonetheless, the total indirect effects of both strategies on group impacts were not significant. A further examination showed that the indirect effects of these mediating variables (network embeddedness and resources) did not differ from each other significantly. Therefore, it is suggested that the total effects of internal strategies on group impacts were primarily ascribed to the direct effects; in contrast, the total effects of external strategies on group impacts were attributed more evenly to direct and indirect effects, even though none was significant.

References

Agarwal, R., and E. Karahanna. 2000. Time flies when you're having fun: Cognitive absorption and beliefs about information technology usage. *MIS Quarterly* 24 (4):665–94.

Aldrich, H. E. 1999. *Organizations evolving*. Thousand Oaks, CA: Sage.

Aldrich, H. E., and J. Pfeffer. 1976. Environments of organizations. *Annual Review of Sociology* 2:79–105.

Ancona, D. G. 1990. Outward bound: Strategies for team survival in an organization. *The Academy of Management Journal* 33 (2):334–65.

Ancona, D. G., and D. F. Caldwell. 1988. Beyond task and maintenance: Defining external functions in groups. *Group and Organization Management* 13 (4):468–94.

Ancona, D. G., and D. F. Caldwell. 1992. Bridging the boundary: External activity and performance in organizational teams. *Administrative Science Quarterly* 37 (4):634–65.

Babchuk, N., and J. N. Edwards. 1965. Voluntary associations and the integration hypothesis. *Sociological Inquiry* 35 (2):149–62.

Bagozzi, R. P., and Y. Youjae. 1988. On the evaluation of structural equation models. *Academy of Marketing Science* 16 (1):74–94.

Baum, J. A. C., and A. V. Shipilov. 2006. Ecological approaches to organizations. In *Sage handbook for organization studies*, ed. S. R. Clegg, C. Hardy, W. Nord, and T. Lawrence, 55–110. London, UK: Sage.

Baum, J. A. C., and J. V. Singh. 1994. Organization-environment coevolution. In *Evolutionary dynamics of organizations*, eds. J. A. C. Baum and J. V. Singh, 379–402. New York, NY: Oxford University Press.

Bimber, B., A. J. Flanagin, and C. Stohl. 2012. *Collective action in organizations: Interaction and engagement and engagement in an era of technological change*. New York, NY: Cambridge University Press.

Calhoun, C. 1986. Computer technology, large-scale social integration, and the local community. *Urban Affairs Review* 22 (2):329–49.

Campbell, D. T. 1965. Variation and selective retention in socio-cultural evolution. In *Social change in developing areas: A reinterpretation of evolutionary theory*, eds. H. R. Barringer, G. I. Blanksten, and R. W. Mack, 19–48. Cambridge, MA: Schenkman.

Chin, W. W. 1998. The partial least squares approach to structural equation modeling. In *Modern methods for business research*, ed. G. A. Marcoulides, 295–336. London, UK: Lawrence Erlbaum Associates.

Chin, W. W., and A. Gopal. 1995. Adoption intention in GSS: Relative importance of beliefs. *The DATA BASE for Advances in Information Systems* 26 (2–3):42–64.

Choi, J. N. 2002. External activities and team effectiveness: Review and theoretical development. *Small Group Research* 33 (2):181–208.

Cohen, J. 1988. *Statistical power analysis for the behavioral sciences*. Hillsdale, NJ: Lawrence Erlbaum Associates.

Coleman, J. S. 1988. Social capital in the creation of human capital. *American Journal of Sociology* 94:95–120.

Dillman, D. A. 2000. *Mail and Internet surveys: The tailored design method* (2nd ed.). New York, NY: John Wiley.

Druskat, V. U., and J. V. Wheeler. 2003. Managing from the boundary: The effective leadership of self-managing work teams. *Academy of Management Journal* 46 (4):435–57.

Faraj, S., and A. Yan. 2009. Boundary work in knowledge teams. *Journal of Applied Psychology* 94 (3):604–17.

Flanagin, A. J., C. Stohl, and B. Bimber. 2006. Modeling the structure of collective action. *Communication Monographs* 73 (1):29–54.

Fornell, C., and D. Larcker. 1981. Evaluating structural equation models with unobservable variables and measurement error. *Journal of Marketing Research* 18 (1):39–50.

Fulk, J., A. J. Flanagin, M. E. Kalman, P. R. Monge, and T. Ryan. 1996. Connective and communal public goods in interactive communication systems. *Communication Theory* 6 (1):60–87.

Fulk, J., R. Heino, A. J. Flanagin, P. R. Monge, and F. Bar. 2004. A test of the individual action model for organizational information commons. *Organization Science* 15 (5):569–85.

Gefen, D., D. W. Straub, and M.-C. Boudreau. 2000. Structural equation modeling and regression: Guidelines for research practice. *Communications of the Association for Information Systems* 4 (7):1–78.

Gotz, O., K. Liehr-Gobbers, and M. Krafft. 2009. Evaluation of structural equation models using the partial least squares (PLS) approach. In *Handbook of partial least squares: Concepts, methods, and applications*, ed. V. E. Vinzi, W. W. Chin, J. Henseler, and H. Wang, 691–711. Berlin, Germany: Springer.

Granovetter, M. S. 1985. Economic action and social structure: The problem of embeddedness. *American Journal of Sociology* 91 (3):481–510.

Guinan, P. J., J. G. Cooprider, and S. Faraj. 1998. Enabling software development team performance during requirements definition: A behavioral versus technical approach. *Information Systems Research* 9 (2):101–25.

Hannan, M. T., and J. H. Freeman. 1977. The population ecology of organizations. *American Journal of Sociology* 82 (5):929–64.

Hansen, M. T. 1999. The search-transfer problem: The role of weak ties in sharing knowledge across organizational subunits. *Administrative Science Quarterly* 44 (1):82–111.

Hayes, A. 2009. Beyond Baron and Kenny: Statistical mediation analysis in the millennium. *Communication Monographs* 76 (4):408–20.

Henseler, J., C. M. Ringle, and R. R. Sinkovics. 2009. The use of partial least squares path modeling in international marketing. *Advances in international marketing* 20 (1):277–320.

Hooghe, M. 2003. Participation in voluntary associations and value indicators: The effect of current and previous participation experiences. *Nonprofit and Voluntary Sector Quarterly* 32 (1):47–69.

Joshi, A., N. Pandey, and G. Han. 2009. Bracketing team boundary spanning: An examination of task-based, team-level, and contextual antecedents. *Journal of Organizational Behavior* 30 (6):731–59.

Katz, J. E., and R. E. Rice. 2002. *Social consequences of Internet use: Access, involvement, and interaction*. Cambridge, MA: MIT Press.

Knoke, D., and D. Prensky. 1984. What relevance do organization theories have for voluntary associations? *Social Science Quarterly* 65 (1):3–20.

Lai, C.-H. 2014a. Can our groups survive? An investigation of the evolution of mixed-mode groups. *Journal of Computer-Mediated Communication* 19 (4):839–854.

Lai, C.-H. 2014b. Understanding the evolution of bona fide mixed-mode groups: An example of Meetup groups. *First Monday* 19 (1). http://firstmonday.org/ojs/index.php/fm/article/view/4681/3810

Laumann, E. O., D. Knoke, and Y. H. Kim. 1985. An organizational approach to state policy formation: a comparative study of energy and health domains. *American Sociological Review* 50 (1):1–19.

Lea, M., and R. Spears. 1991. Computer-mediated communication, de-individuation and group decision-making. *International Journal of Man Machine Studies* 34 (2):283–301.

Lee, S., and P. Monge. 2011. The coevolution of multiplex communication networks in organizational communities. *Journal of Communication* 61 (4):758–779.

Lin, N. 1999. Building a theory of social capital. *Connections* 22 (1):28–51.

Marwell, G., and P. Oliver. 1993. *The critical mass in collective action*. Cambridge, UK: Cambridge University Press.

Mathieu, J., M. T. Maynard, T. Rapp, and L. Gilson. 2008. Team effectiveness. 1997–2007: A review of recent advancements and a glimpse into the future. *Journal of Management* 34 (3):410–476.

McPherson, J. M. 1983. An ecology of affiliation. *American Sociological Review* 48 (4):519–532.

Miner, A. S. 1994. Seeking adaptive advantage: evolutionary theory and managerial action. In *Evolutionary dynamics of organizations*, eds., J. A. C. Baum and J. V. Singh, 76–89. New York, NY: Oxford University Press.

Monge, P. R., and N. S. Contractor. 2003. *Theories of communication networks*. Oxford, UK: Oxford University Press.

Monge, P. R., J. Fulk, M. E. Kalman, A. J. Flanagin, C. Parnassa, and S. Rumsey. 1998. Production of collective action in alliance-based interorganizational communication and information systems. *Organization Science* 9 (3):411–433.

Monge, P. R., B. R. Heise, and D. Margolin. 2008. Communication network evolution in organizational communities. *Communication Theory* 18 (4):449–77.

Nahapiet, J., and S. Ghoshal. 1998. Social capital, intellectual capital, and the organizational advantage. *The Academy of Management Review* 23 (2):242–266.

Nie, N. H. 2001. Sociability, interpersonal relations, and the Internet: Reconciling conflicting findings. *American Behavioral Scientist* 45 (3):419–35.

Oh, H., M. H. Chung, and G. Labianca. 2004. Group social capital and group effectiveness: The role of informal socializing ties. *Academy of Management Journal* 47 (6):860–75.

Olson, M. 1965. *The logic of collective action: Public goods and the theory of groups*. Cambridge, MA: Harvard University Press.

Osei-Bryson, K. M., L. Dong, and O. Ngwenyama. 2008. Exploring managerial factors affecting ERP implementation: An investigation of the Klein-Sorra model using regression splines. *Information Systems Journal* 18 (5):499–527.

Ozcan, P., and K. Eisenhardt. 2009. Origin of portfolios: entrepreneurial firms and strategic action. *Academy of Management Journal* 52 (2):246–79.

Pfeffer, J., and G. R. Salancik. 1978. *The external control of organizations*. New York, NY: Harper & Row.

Postmes, T., and S. Brunsting. 2002. Collective action in the age of the Internet mass communication and online mobilization. *Social Science Computer Review* 20 (3):290–301.

Preacher, K. J., and A. F. Hayes. 2008. Asymptotic and resampling strategies for assessing and comparing indirect effects in multiple mediator models. *Behavior Research Methods* 40 (3):879–91.

Putnam, R. D. 1995. Bowling alone: America's declining social capital. *Journal of Democracy* 6 (1):65–78.

Putnam, R. D. 2000. *Bowling alone: The collapse and revival of American community*. New York, NY: Simon & Schuster.

Rainie, L., K. Purcell, and A. Smith. 2011. The social side of the Internet. Pew Internet & American Life Project. http://www.pewinternet.org/Reports/2011/The-Social-Side-of-the-Internet.aspx (accessed December 2011).

Reagans, R., E. Zuckerman, and B. McEvily. 2004. How to make the team: Social networks vs. demography as criteria for designing effective teams. *Administrative Science Quarterly* 49 (1):101–33.

Rice, R. E., A. E. Grant, J. Schmitz, and J. Torobin. 1990. Individual and network influences on the adoption and perceived outcomes of electronic messaging. *Social Networks* 12 (1):27–55.

Ringle, C., S. Wende, and A. Will. 2005. Smart-PLS Version 2.0 M3. http://www.smartpls.de.

Samuelson, P. A. 1954. The pure theory of public expenditure. *Review of Economics and Statistics* 36 (4):387–389.

Sampson, R. J. 1988. Local friendship ties and community attachment in mass society: A multilevel systemic model. *American Sociological Review* 53 (5):766–79.

Sandler, T., and J. T. Tschirhart. 1980. The economic theory of clubs: An evaluative survey. *Journal of Economic Literature* 18 (4):1481–521.

Sawyer, S., P. J. Guinan, and J. Cooprider. 2010. Social interactions of information systems development teams: a performance perspective. *Information Systems Journal* 20 (1):81–107.

Sessions, L. F. 2010. How offline gatherings affect online communities—when virtual community members "meetup." *Information, Communication & Society* 13 (3):375–95.

Shah, D. V., N. Kwak, and R. L. Holbert. 2001. "Connecting" and "disconnecting" with civic life: Patterns of Internet use and the production of social capital. *Political Communication* 18 (2):141–62.

Shen, C., and C. Cage. 2015. Exodus to the real world? Assessing the impact of offline meetups on community participation and social capital. *New Media & Society* 17 (3):394–414.

Smith, D. H. 1999. The effective grassroots association, part one. Organizational factors that produce internal impacts. *Nonprofit Management and Leadership* 9 (4):443–56.

Smith, D. H. 2000. *Grassroots associations*. Thousand Oaks, CA: Sage.

Sosik, J. J., S. S. Kahai, and M. J. Piovoso. 2009. Silver bullet or voodoo statistics?: A primer for using the partial least squares data analytic technique in group and organizational research. *Group & Organization Management* 34 (1):5–36.

Stolle, D. 2000. Clubs and congregations: The benefits of joining an association. In *Trust in society*, ed. K. S. Cook, 202–44. New York, NY: Russell Sage.

Temme, D., H. Kreis, and L. Hildebrandt. 2010. A comparison of current PLS path modeling software: Features, ease-of-use, and performance. In *Handbook of partial least squares: Concepts, methods and applications*, ed. V. E. Vinzi, W. W. Chin, J. Henseler and H. Wang, 737–56. Berlin, Germany: Springer.

Tenenhaus, M., V. E. Vinzi, Y. Chatelin, and C. Lauro. 2005. PLS path modeling. *Computational Statistics & Data Analysis* 48 (1):159–205.

Tocqueville, A. de. 1968. *Democracy in America*. London, UK: Fontana (Original work published 1835 and 1840).

Tsai, W. 2001. Knowledge transfer in intraorganizational networks: Effects of network position and absorptive capacity on business-unit innovation and performance. *Academy of Management Journal* 44 (5):996–1004.

Turkle, S. 1996. Virtuality and its discontents searching for community in cyberspace. *American Prospect* 24:50–57.

Verba, S., K. L. Schlozman, and H. E. Brady. 1995. *Voice and equality: Civic voluntarism in American politics*. Cambridge, MA: Harvard University Press.

Wasko, M. M., and S. Faraj. 2005. Why should I share? Examining social capital and knowledge contribution in electronic networks of practice. *MIS Quarterly* 29 (1):35–57.

Wasko, M. M., R. Teigland, and S. Faraj. 2009. The provision of online public goods: Examining social structure in an electronic network of practice. *Decision Support Systems* 47 (3):254–65.

Wellman, B., A. Q. Haase, J. Witte, and K. Hampton. 2001. Does the Internet increase, decrease, or supplement social capital? Social networks, participation, and community commitment. *American Behavioralist Scientist* 45 (3):436–55.

Westphal, J. D., S. Boivie, and D. H. M. Chng. 2006. The strategic impetus for social network ties: reconstituting broken CEO friendship ties. *Strategic Management Journal* 27 (5):425–45.

Wetzels, M., G. Odekerken-Schröder, and C. van Oppen. 2009. Using PLS path modeling for assessing hierarchical construct models : Guidelines and empirical illustration. *MIS Quarterly* 33 (1):177–95.

Wollebæk, D., and P. Selle. 2002. Does participation in voluntary associations contribute to social capital? The impact of intensity, scope, and type. *Nonprofit and Voluntary Sector Quarterly* 31 (1):32–61.

Yuan, Y., J. Fulk, M. Shumate, P. R. Monge, J. A. Bryant, and M. Matsaganis. 2005. Individual participation in organizational information commons. *Human Communication Research* 31 (2):212–40.